```
!pip install torch torchvision matplotlib --quiet
```

```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision.transforms as transforms
from torch.utils.data import DataLoader
import matplotlib.pyplot as plt
import os
```

```
!wget http://efrosgans.eecs.berkeley.edu/pix2pix/datasets/facades.tar.gz
!tar -xvf facades.tar.gz
```

```
facades/val/65.jpg
facades/val/66.jpg
facades/val/67.jpg
facades/val/68.jpg
facades/val/69.jpg
facades/val/7.jpg
facades/val/70.jpg
facades/val/71.jpg
facades/val/72.jpg
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facades/val/37.jpg
facades/val/38.jpg
facades/val/39.jpg
facades/val/4.jpg
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facades/val/43.jpg
facades/val/44.jpg
facades/val/45.jpg
facades/val/46.jpg
facades/val/47.jpg
facades/val/48.jpg
facades/val/49.jpg
facades/val/5.jpg
```

```
class Pix2PixDataset(torch.utils.data.Dataset):
    def __init__(self, root_dir, transform=None, limit=200):
        self.root_dir = root_dir
        self.images = os.listdir(root_dir)[:limit]   # LIMIT FOR FAST TRAINING
        self.transform = transform
```

```
    def __len__(self):
        return len(self.images)

    def __getitem__(self, idx):
        img = plt.imread(os.path.join(self.root_dir, self.images[idx]))

        w = img.shape[1]
        input_img = img[:, :w//2]     # facade label
        target_img = img[:, w//2:]    # real building

        input_img = self.transform(input_img)
        target_img = self.transform(target_img)

        return input_img, target_img
```

```
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Resize((256,256)),
    transforms.Normalize((0.5,), (0.5,))
])

train_dataset = Pix2PixDataset("facades/train", transform)
train_loader = DataLoader(train_dataset, batch_size=4, shuffle=True)
```

```
class UNetGenerator(nn.Module):
    def __init__(self):
        super().__init__()

        def down(in_c, out_c):
            return nn.Sequential(
                nn.Conv2d(in_c,out_c,4,2,1),
                nn.BatchNorm2d(out_c),
                nn.LeakyReLU(0.2)
            )

        def up(in_c, out_c):
            return nn.Sequential(
                nn.ConvTranspose2d(in_c,out_c,4,2,1),
                nn.BatchNorm2d(out_c),
                nn.ReLU()
            )

        self.d1 = down(3,64)
        self.d2 = down(64,128)
        self.d3 = down(128,256)

        self.u1 = up(256,128)
        self.u2 = up(256,64)
        self.u3 = nn.ConvTranspose2d(128,3,4,2,1)

        self.tanh = nn.Tanh()

    def forward(self,x):
        d1 = self.d1(x)
        d2 = self.d2(d1)
        d3 = self.d3(d2)

        u1 = self.u1(d3)
        u1 = torch.cat([u1,d2],1)

        u2 = self.u2(u1)
        u2 = torch.cat([u2,d1],1)

        return self.tanh(self.u3(u2))
```

```
class PatchGAN(nn.Module):
    def __init__(self):
        super().__init__()
        self.model = nn.Sequential(
            nn.Conv2d(6,64,4,2,1),
            nn.LeakyReLU(0.2),

            nn.Conv2d(64,128,4,2,1),
            nn.BatchNorm2d(128),
            nn.LeakyReLU(0.2),
```

```
            nn.Conv2d(128,256,4,2,1),
            nn.BatchNorm2d(256),
            nn.LeakyReLU(0.2),

            nn.Conv2d(256,1,4,1,1)
        )

    def forward(self,x,y):
        return self.model(torch.cat([x,y],1))
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

G = UNetGenerator().to(device)
D = PatchGAN().to(device)

criterion_GAN = nn.BCEWithLogitsLoss()
criterion_L1 = nn.L1Loss()

opt_G = optim.Adam(G.parameters(), lr=0.0002)
opt_D = optim.Adam(D.parameters(), lr=0.0002)
```

```
epochs = 5   # FAST TRAINING

for epoch in range(epochs):
    for input_img, target_img in train_loader:

        input_img, target_img = input_img.to(device), target_img.to(device)

        # --- Train D ---
        fake_img = G(input_img)
        D_real = D(input_img, target_img)
        D_fake = D(input_img, fake_img.detach())

        loss_D = (criterion_GAN(D_real, torch.ones_like(D_real)) +
                  criterion_GAN(D_fake, torch.zeros_like(D_fake))) / 2

        opt_D.zero_grad(); loss_D.backward(); opt_D.step()

        # --- Train G ---
        D_fake = D(input_img, fake_img)
        loss_G = criterion_GAN(D_fake, torch.ones_like(D_fake)) + \
                 criterion_L1(fake_img, target_img) * 100

        opt_G.zero_grad(); loss_G.backward(); opt_G.step()

    print(f"Epoch {epoch+1}  Loss D:{loss_D.item():.3f}  Loss G:{loss_G.item():.3f}")
```

```
/usr/local/lib/python3.12/dist-packages/torchvision/transforms/functional.py:154: UserWarning: The given NumPy array is not writ
  img = torch.from_numpy(pic.transpose((2, 0, 1))).contiguous()
Epoch 1  Loss D:0.196  Loss G:48.878
Epoch 2  Loss D:0.076  Loss G:45.748
Epoch 3  Loss D:0.127  Loss G:48.884
Epoch 4  Loss D:0.045  Loss G:50.856
Epoch 5  Loss D:0.380  Loss G:37.326
```

```
input_img, target_img = next(iter(train_loader))
input_img = input_img.to(device)

with torch.no_grad():
    fake_img = G(input_img)

plt.figure(figsize=(12,4))

plt.subplot(1,3,1); plt.title("Input"); plt.imshow(input_img[0].cpu().permute(1,2,0))
plt.subplot(1,3,2); plt.title("Pix2Pix"); plt.imshow(fake_img[0].cpu().permute(1,2,0))
plt.subplot(1,3,3); plt.title("Target"); plt.imshow(target_img[0].permute(1,2,0))
plt.show()
```

```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for inte
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```

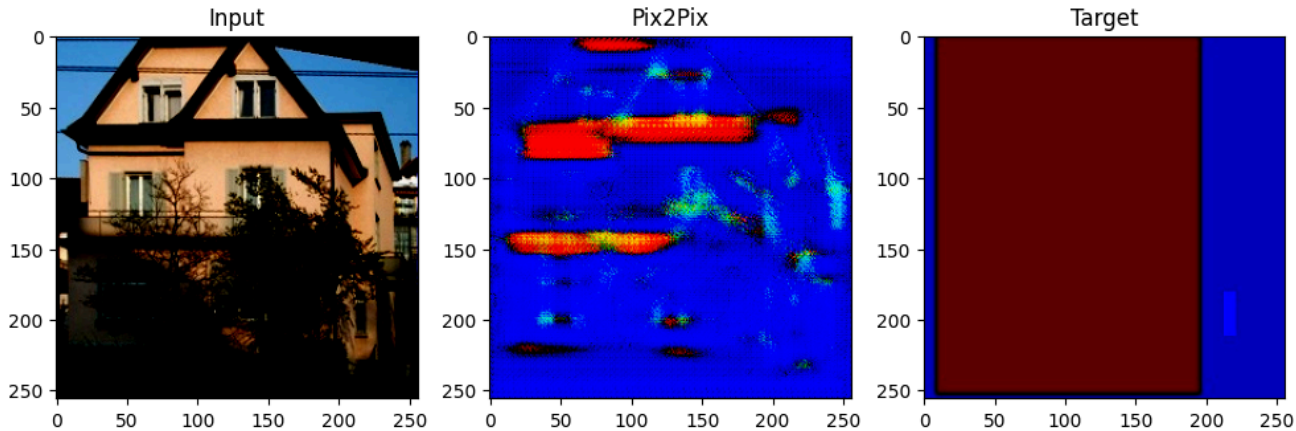


```
class SimpleCNN(nn.Module):
    def __init__(self):
        super().__init__()
        self.encoder = nn.Sequential(
            nn.Conv2d(3,64,4,2,1), nn.ReLU(),
            nn.Conv2d(64,128,4,2,1), nn.ReLU()
        )
        self.decoder = nn.Sequential(
            nn.ConvTranspose2d(128,64,4,2,1), nn.ReLU(),
            nn.ConvTranspose2d(64,3,4,2,1), nn.Tanh()
        )
    def forward(self,x):
        return self.decoder(self.encoder(x))
```

```
cnn = SimpleCNN().to(device)
opt = optim.Adam(cnn.parameters(), lr=0.0002)

for epoch in range(3):
    for x,y in train_loader:
        x,y = x.to(device), y.to(device)
        pred = cnn(x)
        loss = criterion_L1(pred,y)
        opt.zero_grad(); loss.backward(); opt.step()
    print("CNN Epoch", epoch+1, "Loss:", loss.item())
```

```
CNN Epoch 1 Loss: 0.5136864185333252
CNN Epoch 2 Loss: 0.4711779057979584
CNN Epoch 3 Loss: 0.5091410875320435
```